Beining Wu

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RESEARCH SUMMARY

I study **continual learning** in deep networks: where forgetting arises inside a model, which protection strategies provably help, and what a learner should write into its weights versus keep outside them. My work covers **catastrophic forgetting** under parameter-efficient fine-tuning of multimodal LLMs (gradient-subspace protection for MoE-LoRA, IEEE TMM), **federated continual learning** with a layer-wise forgetting decomposition and a provable floor on long-term degradation (ICDCS 2026), **test-time continual learning** for LLM agents (NeurIPS 2026, TTCL; IPCCC 2026), weights-versus-context routing for self-improving agents (TMLR, under review), and **machine unlearning** (ICDCS 2026). I pair formal analysis with controlled multi-seed experiments on 7–13B models, and I wrote an ESI Highly Cited review of continual learning in edge AI (IEEE TNSE).

EDUCATION

South Dakota State University, Brookings, SD, USA — Sep 2024 – Present
PhD in Computer Science, expected May 2028. Advisor: Prof. Jun Huang.

Anhui Normal University, China — Sep 2020 – Jun 2024
B.S. in Mathematics and Applied Mathematics. Coursework: Mathematical Analysis I–III, Real Analysis, Complex Analysis, Advanced Algebra I–II (linear algebra), Probability Theory, Mathematical Statistics, Numerical Methods, Ordinary Differential Equations.

TECHNICAL

Analysis: forgetting bounds under smoothness and heterogeneity assumptions; gradient- and activation-subspace analysis (SVD, orthogonal projection); optimal-stopping and budgeted-allocation formulations.

ML: Python, PyTorch, Hugging Face; LoRA and MoE-LoRA fine-tuning of 7–13B vision-language models (LLaVA-1.5, Qwen2.5-VL); continual-learning benchmarks (CoIN-6, CoIN-Long-10) against fourteen baselines; GUI agents on AndroidWorld and MobileGym; federated learning and machine unlearning.

Compute: A100/H100 clusters; NSF ACCESS Discover allocation as PI (750,000 credits); NVIDIA Jetson and rover testbed; controlled multi-seed experiments with 95% confidence intervals; Linux, Git.

PUBLICATIONS

Google Scholar (October 2026): 784 citations, h-index 13, i10-index 17. One ESI Hot Paper (Top 0.1%) and two ESI Highly Cited Papers (Top 1%).

Continual Learning and Forgetting in Deep Networks

- **Beining Wu**, Zihao Ding, Jun Huang. "Exposing and Resolving Spurious Isolation in Federated Multimodal Continual Learning." *IEEE Transactions on Multimedia*, 2026. DOI: 10.1109/TMM.2026.3736516. arXiv:2605.01061 • Code.

  Mixture-of-experts routing does not isolate tasks: we prove that gradient conflict inside an active expert survives every non-negative routing, and that under a per-expert rank budget the activation subspaces of two tasks must overlap (a dimension-counting bound); measured activation overlap is about 4× the gradient overlap. POLARIS protects a per-expert gradient subspace whose orthogonality survives federated averaging with bounded truncation loss, and beats fourteen baselines on LLaVA-1.5-7B/13B and Qwen2.5-VL-7B; the margin grows from +3.16 pp on CoIN-6 to +6.00 pp on CoIN-Long-10.

- **Beining Wu**, Jun Huang, Yanxiao Zhao. "From Alpha to Omega: Lifecycle-Aware Forgetting Defense in Federated Continual Learning for Planetary Exploration." *IEEE International Conference on Distributed Computing Systems (ICDCS)*, 2026. (Acceptance rate 18.59%.) IEEE Xplore.

  Forgetting decomposes layer by layer, so uniform rehearsal is provably suboptimal at a matched protection budget; even a perfect training-time defense leaves a long-term degradation floor from data heterogeneity and buffer dilution, which motivates a separate recovery step with bounded sample cost. +2.3 to 3.3% mIoU on three Mars terrain datasets, deployed on a heterogeneous rover testbed.

- **Beining Wu**, Jun Huang. "Lifecycle-Aware Federated Continual Learning in Mobile Autonomous Systems." *Under review, IEEE Transactions on Mobile Computing*, 2026. arXiv:2604.20745.

  Journal version of the dual-timescale analysis: layer-selective rehearsal plus rapid recovery reach up to 8.3% mIoU over the strongest federated baseline and 31.7% over fine-tuning.

- **Beining Wu**, Zihao Ding, Jun Huang. "A Review of Continual Learning in Edge AI." *IEEE Transactions on Network Science and Engineering*, 2026. **ESI Highly Cited Paper (Top 1%).**

Continual Learning for LLM Agents: Weights, Context, and Memory

- **Beining Wu**, Zihao Ding, Jun Huang. "Not All Experience Belongs in the Weights: Component Routing for Self-Improving GUI Agents." *Under review, Transactions on Machine Learning Research (TMLR)*, 2026. arXiv:2610.01787.

  A GUI agent's trajectory bundles locators, procedures, state facts, and lessons, and they belong in different places: locators and lessons gain

when fine-tuned into the weights, procedures and state facts when retrieved into the context. A rule on two properties measured before training predicts the destination for a held-out model family in 24 of 24 cells, and routing by it beats the better single destination by 3.5 points (MobileGym and AndroidWorld, three 7–8B model families).

- **Beining Wu**, Zihao Ding, Jun Huang. "ERRAND: Budgeted Maintenance of Agent Memory." ***NeurIPS 2026**, Towards Test-Time Continual Learning Agents*. arXiv:2609.29545.

  A recheck competes with the task it protects for the same actions and is funded only when resolving a doubt is worth the going rate. At equal action budgets in two drifting tool-use worlds it leads eager revalidation by 10.0 points; with no cap it stops at 11.0% of steps, while eager revalidation spends 70.7% and still trails by 4.5 points.

- **Beining Wu**, Jun Huang. "CrystalMem: Elastic Memory for Self-Evolving LLM Agents via Knowledge Crystallization." *Under review, IEEE Transactions on Cloud Computing*, 2026. arXiv:2608.00303.

  Squeeze an agent's memory budget and restore it, and capability does not return; we prove that any keep-or-drop policy leaves a residual deficit, and fidelity states with verified recovery remove it (+4.6 pp at equal budget, seven environments, six backbones).

- **Beining Wu**, Zihao Ding, Jun Huang, Yanxiao Zhao. "Forget to Improve: On-Device LLM-Agent Continual Learning via Budget-Curated Memory." *IEEE International Performance, Computing, and Communications Conference (IPCCC)*, 2026. arXiv:2606.25115.

  One score decides which experiences an agent keeps, shares with peers, and accepts from them; memory drops 2.7× and memory-injection success falls from 0.75 to 0 on four rovers and three Jetson nodes.

- **Beining Wu**, Jun Huang, Yanxiao Zhao. "REBASE: Device-Cloud Experience Coherence for GUI Agents Across App Updates." *Under review, IEEE International Conference on Communications (ICC)*, 2027. arXiv:2609.32130.
- **Beining Wu** (as Michael Wu), Arquimedes Canedo. "Invalidation Contracts for Cross-Episode Agent Memory." *arXiv preprint*, 2026. arXiv:2609.00243 • Live demo. Joint work with Siemens.

Machine Unlearning and Federated Learning

- Zihao Ding, **Beining Wu**, Jun Huang. "SCALE: Sensitivity-Aware Federated Unlearning with Information Freshness Optimization for Mobile Edge Computing." *IEEE International Conference on Distributed Computing Systems (ICDCS)*, 2026. (Acceptance rate 18.59%.) IEEE Xplore • arXiv:2605.22589.
- Zihao Ding, **Beining Wu**, Jun Huang. "EASE: Federated Multimodal Unlearning via Entanglement-Aware Anchor Closure." *Under review, International Conference on Learning Representations (ICLR)*, 2027. arXiv:2605.00733.
- **Beining Wu**, Zihao Ding, Jun Huang. "RELIEF: Turning Missing Modalities into Training Acceleration for Federated Learning on Heterogeneous IoT Edge." *IEEE Internet of Things Journal*, 2026. DOI: 10.1109/JIOT.2026.3725593. arXiv:2604.04243.
- **Beining Wu**, Jun Huang, Qiang Duan, Liang Dong, Zhipeng Cai. "Enhancing Vehicular Platooning with Wireless Federated Learning: A Resource-Aware Control Framework." *IEEE/ACM Transactions on Networking*, 2025.
- Jun Huang, **Beining Wu**, Qiang Duan, Liang Dong, Shui Yu. "A Fast UAV Trajectory Planning Framework in RIS-assisted Communication Systems with Accelerated Learning via Multithreading and Federating." *IEEE Transactions on Mobile Computing*, 2025. **ESI Hot Paper (Top 0.1% in Computer Science).**
- **Beining Wu**, Jun Huang, Qiang Duan. "Real-time Intelligent Healthcare Enabled by Federated Digital Twins with AoI Optimization." *IEEE Network*, 2025. **ESI Highly Cited Paper (Top 1%).**

RESEARCH EXPERIENCE

Graduate Research Assistant — South Dakota State University, Brookings, SD May 2024 – Present

NSF CNS-2348422: machine learning for resource-constrained autonomous systems. Supervisor: Prof. Jun Huang.

- Lead a research line on continual learning in deep networks: formal analysis of where forgetting arises (layer-wise forgetting decomposition, gradient conflict inside MoE-LoRA experts, subspace-entanglement bounds under PEFT) and methods validated on 7–13B multimodal models and on a federated rover fleet (IEEE TMM 2026; ICDCS 2026).
- Extended the line to LLM agents that keep learning after deployment: which parts of an agent's experience to fine-tune into the weights and which to keep in context, how stored experience stays correct as apps change, and when a possibly stale memory is worth rechecking under a fixed action budget (NeurIPS 2026, TTCL; TMLR under review).

Research Collaborator — Siemens (remote), with Dr. Arquimedes Canedo Aug 2026 – Present

- Co-designed invalidation contracts for cached LLM-agent memory; built the evaluation harness that runs seven LLM backends over three serving paths and two domains (about 9,400 episodes) and reports token cost, eviction precision, and first-try success; released the public demo on Hugging Face Spaces and first-authored the paper (arXiv:2609.00243).

PROFESSIONAL SERVICE

TPC Member, IEEE ICNC 2026, 2027; Reviewer, ICLR 2026, NeurIPS 2026 IAB Workshop, IEEE/RSJ IROS 2026, and WACV 2027 LVR Workshop; Book Proposal Reviewer, Springer Nature (Computer Science), 2026; journal reviewer for 18 venues (IEEE JSAC, TMC, TDSC, IoTJ, and others).